

1) Select K Best Regression Algorithm

K=4

	Linear	SVMI	SVMnl	Decision	Random
ChiSquare	0.304963	0.256858	0.430795	0.479167	0.599392

K=5

	Linear	SVMI	SVMnl	Decision	Random
ChiSquare	0.551985	0.545395	0.749654	0.696181	0.836806

K=6

	Linear	SVMI	SVMnl	Decision	Random
ChiSquare	0.599041	0.586446	0.838962	0.869792	0.897569

K=7

	Linear	SVMI	SVMnl	Decision	Random
ChiSquare	0.657035	0.641906	0.893007	0.826389	0.916233

K=8

	Linear	SVMI	SVMnl	Decision	Random
ChiSquare	0.646457	0.612199	0.891274	0.869792	0.898872

K=7 is best feature selection

2) Select K Best Classification Algorithm

K=6

	Logistic	SVMl	SVMnl	KNN	Navie	Decision	Random
ChiSquare	0.95	0.96	0.96	0.93	0.89	0.97	0.97

K=7

	Logistic	SVMl	SVMnl	KNN	Navie	Decision	Random
ChiSquare	0.97	0.97	0.97	0.98	0.91	0.95	0.96

K=8

	Logistic	SVMl	SVMnl	KNN	Navie	Decision	Random
ChiSquare	0.97	0.97	0.97	0.98	0.91	0.95	0.96

K=9

	Logistic	SVMl	SVMnl	KNN	Navie	Decision	Random
ChiSquare	0.97	0.97	0.98	0.99	0.87	0.95	0.98

K=10

	Logistic	SVMl	SVMnl	KNN	Navie	Decision	Random
ChiSquare	0.99	0.98	1.0	0.99	0.92	0.96	0.97

K=11

	Logistic	SVMl	SVMnl	KNN	Navie	Decision	Random
ChiSquare	0.99	0.98	0.99	0.98	0.92	0.97	0.99

K=10 and K=11 is best feature selection

RFE Regression:

N=8

	Linear	Ridge	Lasso	Decision	SVR	Random
Linear	0.709204	0.709165	0.599767	0.952168	0.684292	0.939159
Ridge	0.709204	0.709165	0.599767	0.952168	0.684292	0.939159
Lasso	0.604073	0.604097	0.538829	0.826389	0.562445	0.837778
Decision	0.703917	0.704033	0.579447	0.782986	0.673437	0.931055
SVR	0.701052	0.701088	0.599767	0.82978	0.679964	0.930577
Random	0.701652	0.701497	0.573936	0.782986	0.679374	0.915178

N=9

	Linear	Ridge	Lasso	Decision	SVR	Random
Linear	0.716216	0.71625	0.599767	0.968654	0.684977	0.962849
Ridge	0.716216	0.71625	0.599767	0.968654	0.684977	0.962849
Lasso	0.60317	0.603203	0.538829	0.869792	0.564522	0.835694
Decision	0.702878	0.703025	0.579447	0.826389	0.672776	0.92783
SVR	0.700809	0.700773	0.599767	0.82978	0.682756	0.927438
Random	0.715774	0.715834	0.581397	0.782986	0.681605	0.912287

N=9 is best feature selection

RFE Classification:

N=8

	Logistic	SVMI	SVMnl	KNN	Navie	Decision	Random	GradientBoosting
Logistic	0.98	0.98	0.98	0.96	0.98	0.95	0.99	1.0
LinearSVC	0.99	0.99	0.99	0.98	0.98	0.99	0.98	0.99
DecisionTree	0.98	0.97	0.98	0.99	0.94	0.99	0.99	0.99
RandomForest	0.98	0.98	0.99	0.98	0.92	0.96	0.97	0.97
GradientBoosting	0.97	0.97	0.98	0.98	0.94	0.96	0.99	0.99

N=9

	Logistic	SVMI	SVMnl	KNN	Navie	Decision	Random	GradientBoosting
Logistic	0.99	0.99	0.99	0.99	0.98	0.95	1.0	0.99
LinearSVC	0.98	0.98	0.99	0.99	0.98	0.99	0.98	0.99
DecisionTree	0.98	0.97	0.98	0.99	0.94	0.99	1.0	1.0
RandomForest	0.99	0.97	1.0	0.99	0.92	0.96	0.99	0.99
GradientBoosting	0.98	0.98	0.99	0.98	0.95	0.95	0.98	1.0

3. Feature Importance regression:

N=8

	Linear	Ridge	Lasso	Decision	SVR	Random
Linear	0.709204	0.709165	0.599769	0.952168	0.684292	0.941411
Ridge	0.709204	0.709165	0.599769	0.952168	0.684292	0.941411
Lasso	0.604073	0.604097	0.538828	0.782986	0.562445	0.833485
Decision	0.680503	0.681129	0.58153	0.739583	0.632372	0.849484
SVR	0.59035	0.59066	0.526866	0.827847	0.573233	0.824178
Random	0.704117	0.704329	0.581403	0.782986	0.664507	0.891432

N=9

	Linear	Ridge	Lasso	Decision	SVR	Random
Linear	0.716216	0.71625	0.599769	0.968654	0.684977	0.963534
Ridge	0.716216	0.71625	0.599769	0.968654	0.684977	0.963534
Lasso	0.60317	0.603203	0.538828	0.869792	0.564522	0.835703
Decision	0.680659	0.681204	0.581409	0.826389	0.629183	0.891563
SVR	0.667347	0.667502	0.594996	0.96718	0.648974	0.938206
Random	0.715774	0.715834	0.581403	0.782986	0.681605	0.90622

N=10

	Linear	Ridge	Lasso	Decision	SVR	Random
Linear	0.714887	0.714957	0.599769	0.968654	0.690464	0.96383
Ridge	0.714887	0.714957	0.599769	0.968654	0.690464	0.96383
Lasso	0.60303	0.603111	0.538828	0.826389	0.556655	0.84138
Decision	0.696777	0.697212	0.581409	0.782986	0.653055	0.910326
SVR	0.662764	0.66281	0.594996	0.923777	0.643035	0.942185
Random	0.713361	0.713391	0.581403	0.826389	0.680718	0.915686

N=10 is best feature selection

Feature Importance classification:

N=8

	Logistic	SVMI	SVMnl	KNN	Navie	Decision	Random	GradientBoosting
Logistic	0.98	0.98	0.98	0.96	0.98	0.97	1.0	1.0
LinearSVC	0.96	0.96	0.95	0.96	0.95	0.95	0.97	1.0
DecisionTree	1.0	1.0	0.98	0.99	0.93	0.96	0.96	0.99
RandomForest	0.98	0.98	0.99	0.98	0.9	0.97	0.98	0.98
GradientBoosting	0.97	0.97	0.98	0.98	0.94	0.98	0.96	1.0

N=9

	Logistic	SVMl	SVMnl	KNN	Navie	Decision	Random	GradientBoosting
Logistic	0.99	0.98	0.98	0.95	0.98	0.95	1.0	1.0
LinearSVC	0.98	0.98	0.98	0.97	0.98	0.98	1.0	1.0
DecisionTree	0.98	1.0	0.97	0.97	0.93	0.96	0.99	0.99
RandomForest	0.99	0.97	1.0	1.0	0.93	0.96	1.0	0.99
GradientBoosting	0.98	0.98	0.99	0.98	0.95	0.98	0.99	1.0

N=10

	Logistic	SVMl	SVMnl	KNN	Navie	Decision	Random	GradientBoosting
Logistic	0.98	0.97	0.98	0.97	0.98	0.96	0.99	1.0
LinearSVC	0.98	0.97	0.98	0.97	0.98	0.95	0.99	1.0
DecisionTree	0.97	0.97	0.96	0.98	0.93	0.96	1.0	0.99
RandomForest	1.0	0.97	1.0	1.0	0.96	0.96	0.98	1.0
GradientBoosting	0.98	0.99	0.99	0.99	0.98	0.95	0.99	1.0

N=11

	Logistic	SVMl	SVMnl	KNN	Navie	Decision	Random	GradientBoosting
Logistic	0.99	0.98	0.99	0.98	0.98	0.96	1.0	1.0
LinearSVC	0.98	0.97	0.98	0.98	0.98	0.96	1.0	1.0
DecisionTree	0.98	0.97	0.96	0.97	0.94	0.95	1.0	1.0
RandomForest	1.0	0.97	1.0	1.0	0.96	0.96	0.98	0.98
GradientBoosting	0.98	0.99	0.99	0.99	0.98	0.95	0.99	1.0

N=11 is best feature selection

PCA Classification:

N=9

	Logistic	SVMI	SVMnl	KNN	Navie	Decision	Random
PCA	0.99	0.98	0.99	0.96	0.98	0.99	0.99

N=10

	Logistic	SVMI	SVMnl	KNN	Navie	Decision	Random
PCA	0.99	0.98	0.99	0.97	0.98	0.99	1.0

N=11

	Logistic	SVMI	SVMnl	KNN	Navie	Decision	Random
PCA	0.99	0.98	0.98	0.97	0.98	0.99	1.0

N=10 is best feature selection

PCA Regression:

N=8

	Linear	SVMI	SVMnl	Decision	Random
PCA	0.667804	0.618957	0.914228	1.0	0.990885

N=9

	Linear	SVMI	SVMnl	Decision	Random
PCA	0.675469	0.618939	0.911755	0.956597	0.986111

N=10

	Linear	SVMl	SVMnl	Decision	Random
PCA	0.671568	0.628273	0.920411	0.956597	0.986111

N=8 is best feature selection